

Analysts are encouraged to duplicate these simulation tests with different order sizes, volatility, liquidity, and market impact sensitivity to observe the difficulty with calibrating a stock specific model. Then repeat the same experiment for a group of stocks, say 100 plus stocks, and then again with say 500 stocks. A pattern will start to emerge.

Simulation Scenario

Stock RLK. Current price $P_0 = \$50$, annualized volatility $\sigma = 30\%$, the stock trades 1,000,000 shares per day. The customer order will consist of order sizes from -25% to $+25\%$ ADV. The average trade size is 200 shares per trade and the distribution of trade size is shown below as $Shares(t)$.

Simulate market trades for the customer and all other investors. Let the total volume traded from all other market participants be equal to one million shares where the side of the order is randomly assigned (50% chance of a buy and 50% chance of a sell), and let the customer order size and side be specified in advance. Then the simulated data is as follows. Let,

$$P_0 = 50$$

$$Side(t) = \begin{cases} +1 & 0.50 \\ -1 & 0.50 \end{cases}$$

$$Customer\ Side(t) = +1 \text{ (buy order)}$$

The customer order side is specified in advance:

$$Shares(t) = \begin{cases} 0.90 & 100 \\ 0.06 & 500 \\ 0.03 & 1000 \\ 0.01 & 5000 \end{cases}$$

$$Avg\ Share\ Size = 200$$

$$Number\ trades\ during\ the\ day = 5000$$

$$ADV = 1,000,000$$

$$MI(Shares(t)) = 0.0000025 \cdot Shares(t)$$

For a 20% order size on a stock that trades one million shares per day, this market impact cost will be equivalent to 40 bp (and is consistent with our findings in Chapter 5). For simplicity, we assume a linear impact relationship but analysts are encouraged to try various formulations of the impact model.